Assignment 2: Constraint Satisfaction Problem (CSP) for Timetable Scheduling

Objective:
You will model and solve a CSP by generating a timetable for Semester 2 for 1CS students (3 groups). You will implement all algorithms from scratch (no CSP libraries), generate a **timetable** that satisfies hard constraints, and optionally satisfy soft constraints for extra credit.

Important Instructions:

- Work in groups of two students from the same group.
- The project will be presented in person; no online submission is required.
- During evaluation, you must explain your code and algorithms.

Week Structure:

- Sunday: 5 slots
- Monday: 5 slots
- Tuesday: 3 slots (morning only)
- Wednesday: 5 slots
- Thursday: 5 slots
  Each slot can be represented as (Day, Slot Number)

Courses and Sessions:

- Sécurité: 1 Lecture + 1 TD (Teacher 1)
- Méthodes Formelles: 1 Lecture + 1 TD (Teacher 2)
- Analyse Numérique: 1 Lecture + 1 TD (Teacher 3)
- Entrepreneuriat: 1 Lecture (Teacher 4)
- Recherche Opérationnelle 2: 1 Lecture + 1 TD (Teacher 5)
- Distributed Architecture & Intensive Computing: 1 Lecture + 1 TD (Teacher 6)
- Réseaux 2: 1 Lecture + 1 TD + 1 TP (Teacher 7 + Teachers 8-9)
- Artificial Intelligence: 1 Lecture + 1 TD + 1 TP (Teacher 10 + Teachers 11-12)

CSP Formulation:

- Variables: Each course session is a variable.
- Domains: All available slots in the week.
- Constraints:
  Hard Constraints:
  1. No two sessions for the same group at the same time.
  2. A teacher cannot teach two sessions at the same time.
  3. Maximum 3 consecutive sessions per day.

Required Algorithms:

- Backtracking Search
- AC-3 algorithm (Arc Consistency)
- MRV (Minimum Remaining Values) and one additional heuristic (Degree or LCV)
  All algorithms must be implemented from scratch.

Timetable Visualization:

- Display timetable clearly: table, colored schedule, simple interface, or well-formatted console output.

Evaluation
Mandatory Work (**/13**):

- CSP formulation
- Domain modeling
- Hard constraints
- Backtracking algorithm
- AC-3 implementation
- Heuristics implementation
- Correct timetable generation
- Timetable visualization

Extra Credit (/2 → total /15):

- Soft constraint (teacher max two workdays)

Deadline:

- To be defined by TP teacher.